Common difference / ratio and explain all year

A mathematical statement has a field and assumptions. Before demonstrating it, write them; to refute it, give an example that satisfies the assumptions and contradicts the conclusion. or simulation helps to search but does not prove a universal statement.

Tool	Task and production	Accurate review
Membership and ensemble	Describe a right as a set of points; check all coordinates.	Implementation 1, p. 132.
Quantifiers and negation	Denying universal positivity and giving A witness rank.	R10; application 2.
Implications and converse	Distinguish fixed point and convergence; give a diverging sequence.	Implementation 4.
Equation	Solve a logarithmic equation After fixing the domain.	Implementation 1.
Mathematical induction	Identify property, initialize and prove Heredity to an arbitrary rank.	Applications 3 then 5.
Counter-example	Remove a hypothesis from the VIT; show this that fails.	Implementation 5.
Contrapositive	Show that two vectors of non-product None of them are orthogonal.	R60 and R61.
Absurd	Suppose four tops of a tetrahedron not flattened in a plane.	Unit 10.2.
Case breakdown	Process a null parameter before dividing.	Implementation 5.
Adequate condition	Distinguish a size guaranteed by a limit of an exact minimum size.	Applications 4 and 5.

Reusable oral protocol

In a minute: announce the object studied and the conclusion, name the tool and its assumptions, give the decisive step, then a check. The partner listens without interrupting and reformulates the conclusion ; A question then asks why a hypothesis is necessary. in each chapter are used as supports.

Short return of the listener

Is the object defined? Is the decisive hypothesis named? Is the link between the steps is the result interpreted? Choose a single point to improve before Do it again.

Oral progression

September: read a clue and say the induction hypothesis. November: compare two limits December: distinguish existence and uniqueness, then two conditionings. February: describe a projected and a continuous evolution. March: interpret the unity of an integral. April: distinguishing probabilistic guarantee and certainty. May: expose a complete problem and This short time complements the O1 to O3 workshops.

Results accepted and demonstrations to be reused

Operations and boundary composition, monotonous convergence, VIT, equivalences
general convexity, the general existence of antiderivatives and the additivity of variance under independence
are accepted in the framework indicated in progress. Evidence from the register is demonstrated and
A demonstration of ITV by dichotomy, the method of
Newton, oblique asymptotes and repeat combinations are deepenings
possible: their knowledge is not a hidden prerequisite of the proposed evaluations.

TP supplement: compare samples

The program below extends P5·P6. The tolerance in counting is only used to absorb
decimal representation of the floats; the exact calculation uses rational fractions.

```
"""Compare averages of several independent samples."""
From random    import    Random
From Statistics          import     mean, pstdev
From math    import     sqrt, comb
From fractions        import     Fraction

def   experience(n, repetitions=1000, p=0.5, seed=2026):
      if    n  <=  0 gold Repetitions < 2           or not      0  <=  p  <=  1:
            raise    ValueError(       "Out-of-domain settings"                  )
      rng = Random(grain)
      mean = [           sum(rng.random() < p           for     _ in rows       (n))/n
                         for    _ in rows       (repetitions)]
      return     {"n":  n, "average"       : average,
                  "ecart_type_observe"           : pstdev(average),
                  "ecart_type_theoretical"           : sqrt(p*(1-p)/n),
                  "proportion_ecart_at least_0_1"                 :
                         sum (abs  (m-p)  >=  0.1-1e-12 for     m in    average)/repetitions}

def   ecart_exact_equilibrium(n, delta):
      """delta is a Fraction: the entire boundaries are accurate."""
      if    n  <=  0 gold delta <= 0:
            raise    ValueError(       "n and delta must be positive"                        )
      favourable =        sum  (comb(n,k)      for     k in rows       (n+1)
                              if abs     (Fraction(k,n)-Fraction(1.2)) >= delta)
      return    Fraction(favourable,2**n)

if   __name__==       "__main__"      :
      import   json
      print   (json.dumps([experience(n))              for     n in    (10,100,1000)],indent=2))
```